

The divisor ladder at $t = -1$ is an intertwiner, not a power
 $R_4(-1) \neq c R_2(-1)^2$, and the relation that replaces it is rigid: it holds at $t = -1$ and
 nowhere else

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Abstract

Let $R_e(t)$ be the operator on symmetric functions that adds a connected e -ribbon with weight t^{ht} . The browse log of 2026-09-09 (evening) ranked, as the cheapest decisive test available, the relation

$$R_4(-1) \doteq R_2(-1)^2,$$

obtained by transcribing a *divisor ladder* for Hall–Littlewood functions at roots of unity into the operator picture. We decide it, and we replace it.

(A) The relation is false, and false maximally. $R_4(-1) \neq c R_2(-1)^2$ for every scalar c (Theorem 4); indeed the operators $\{R_e(-1)\}_{e \geq 1}$ are algebraically independent, so *no* polynomial relation of any shape holds among them (Corollary 5). A divisor ladder is not merely unproved here; there is no room for one.

(B) The divisor structure is real, but it is a conjugation. Let $\psi^e : \Lambda \rightarrow \Lambda$ be the Adams operation $p_m \mapsto p_{em}$ (plethysm with p_e). Then for all $d, e \geq 1$

$$R_{de}(-1) \circ \psi^e = \psi^e \circ R_d(-1).$$

The divisor $d \mid de$ is implemented by an *intertwiner*, not by an e -th power (Theorem 10).

(C) The intertwiner is rigid at the anchor. For $e \geq 2$ and $de \geq 2$, the equation $R_{de}(u) \circ \psi^e = c \psi^e \circ R_d(t)$ with $c \neq 0$ forces $u = -1$, $c = 1$, and (when $d \geq 2$) $t = -1$ (Theorem 12). This is a new characterisation of $t = -1$, independent in content from the Q75 characterisation (“ $R_e(t)$ is a multiplication operator iff $t = -1$ ”): the obstruction is a single power-sum coefficient, $[p_{1^n}] f_n(t) = (1+t)^{n-1}/n!$.

(D) The sharp diagnosis. ψ^e divides ribbon size by e . Under ψ^2 , a 4-ribbon corresponds to a domino and *two* dominoes correspond to *two* single boxes. So the conjecture $R_4(-1) = c R_2(-1)^2$ descends, along the intertwiner and through the injectivity of ψ^2 , to $p_2 = c p_1^2$ (Proposition 15). The transcription confused “one e -ribbon upstairs” with “ e ribbons”.

(E) A by-product for the exponent contest. At $t = -1$ the proved cocycle $\omega R_e(t) \omega = t^{e-1} R_e(1/t)$ specialises to the classical $\omega(p_e) = (-1)^{e-1} p_e$. This is an *untuned* confirmation of the exponent $e-1$, and it separates t^{e-1} from the competing t^2 at every *even* e — including the $e = 4$ the brief demanded — subject to a category-check on the rival that I could not perform in this session and state explicitly as a gap (§8).

1 What was asked, what I ran instead, and why

The brief `state/PROVE.md` (2026-09-09 cycle 2) set Q121 — decide whether Zabrocki’s flip $\hat{V}$ acts on $R_e(t)$ by a scalar cocycle — and opened with a mandatory gate: *read today’s browse log first, and if it re-ranks, follow the log and say so.*

It re-ranks. The evening browse log (`memory/reading/2026-09-09-evening.md`, written 16:17, after the brief) closes with a revised ranking whose first item is not Q121 but Q130:

“**Q130** — **test** $R_4(-1) \doteq R_2(-1)^2$. Disk-only, at a point I already compute, and the first test that uses $e = 4$ ’s divisor interior. Cheapest decisive thing on the list.”

The log also *retires* the brief’s mandatory negative control (“**Not to run:** the six-sightings-at- $t = 2$ control (Q122). Its premise is refuted”) before conditionally reinstating it at rank 4 behind a re-derivation I have not done. I therefore did not run the $t = 2$ control; §10 records this as a deliberate omission with its reason.

So this session ran Q130. Recording the override is, per the gate, a result and not a deviation. Two further items of the brief turned out to be stale in the same way and are recorded in §10.

Where Q130 came from, and what this paper does *not* claim. Q130 was generated by transcribing a divisor ladder for Hall–Littlewood functions at roots of unity,

$$Q'_{(r^l)}(x; \zeta_d) = (Q'_{(r^d)}(x; \zeta_d))^{l/d} \quad (d \mid l),$$

recorded in my reading log as Descouens–Morita, `math/0611522`, Prop. 6.1. That entry sits at extraction level `deep-read`, but the read was an *agent’s*, not mine. **Nothing below uses that proposition.** All statements proved here are statements about my own operators $R_e(t)$, and what §7 refutes is the *operator transcription*, not the source. The source theorem, as recorded, is a statement about a family of *vectors* indexed by rectangles, in a setting where the root order and the ribbon size are locked equal; $R_4(-1)$ is a 4-ribbon operator evaluated at a *second* root of unity, and so lies outside that locked pair. Whether the source statement is correctly recorded is not decided here and is not needed.

2 Conventions

Convention 1. Λ is the ring of symmetric functions over $\mathbb{Q}(t)$, with Schur basis $\{s_\lambda\}$, power sums p_m , and Hall inner product $\langle p_\lambda, p_\mu \rangle = z_\lambda \delta_{\lambda\mu}$, $z_\lambda = \prod_i i^{m_i} m_i!$. For $e \geq 1$,

$$R_e(t) s_\mu = \sum_{\lambda} t^{\text{ht}(\lambda/\mu)} s_\lambda,$$

the sum over $\lambda \supset \mu$ with λ/μ a connected border strip (e -ribbon) of size e , and $\text{ht} = (\#\text{rows}) - 1$. This is Convention 2.1 of `2026-09-07-Q92-cross-rank-commutator.tex`, unchanged. We write M_f for multiplication by $f \in \Lambda$, $f^\perp$ for its adjoint, and ω for the standard involution $s_\lambda \mapsto s_{\lambda'}$.

We write

$$f_n(t) := R_n(t) \cdot 1 = \sum_{k=0}^{n-1} t^k s_{(n-k, 1^k)},$$

the notation of `2026-09-03-Q75-multiplication-defect.tex`; the equality holds because the border strips of size n inside $\lambda \supset \emptyset$ are exactly the hooks, the hook $(n - k, 1^k)$ having height k .

Convention 2 (the Adams operation). $\psi^e : \Lambda \rightarrow \Lambda$ is the $\mathbb{Q}(t)$ -algebra endomorphism determined by $\psi^e(p_m) = p_{em}$. Equivalently $\psi^e(F) = p_e \circ F = F[p_e]$, plethysm with p_e , i.e. $F(x_1, x_2, \dots) \mapsto F(x_1^e, x_2^e, \dots)$. It is an injective ring homomorphism with image $\mathbb{Q}(t)[p_e, p_{2e}, p_{3e}, \dots]$, and it multiplies degrees by e .

We use throughout the following, which is Theorem B(i) of 2026-09-03-Q75-multiplication-defect.tex and is **proved**:

Theorem 3 (Q75; equivalently the Murnaghan–Nakayama rule). $R_e(-1) = M_{p_e}$ for every $e \geq 1$.

3 The refutation, in its maximal form

Theorem 4. *There is no scalar c with $R_4(-1) = c R_2(-1)^2$.*

Proof. By Theorem 3, $R_4(-1) = M_{p_4}$ and $R_2(-1)^2 = M_{p_2}^2 = M_{p_2^2}$. Applying both to 1 gives p_4 and p_2^2 . In the Schur basis

$$p_4 = s_{(4)} - s_{(3,1)} + s_{(2,1,1)} - s_{(1^4)}, \quad p_2^2 = s_{(4)} - s_{(3,1)} + 2s_{(2,2)} - s_{(2,1,1)} + s_{(1^4)}.$$

The coefficient of $s_{(2,2)}$ is 0 on the left and 2 on the right, forcing $c = 0$; the coefficient of $s_{(4)}$ is 1 on both sides, forcing $c = 1$. Contradiction. $\square$

The refutation is not marginal. It is structural, and the structure kills every relation of that kind at once:

Corollary 5. *The operators $\{R_e(-1)\}_{e \geq 1}$ are algebraically independent over $\mathbb{Q}(t)$: the subalgebra of $\text{End}_{\mathbb{Q}(t)}(\Lambda)$ they generate is a polynomial algebra on countably many generators. In particular, for all $d, e \geq 1$ with $e \geq 2$ and $d \geq 1$, and every scalar c , $R_{de}(-1) \neq c R_d(-1)^e$ unless $d = de$, i.e. unless $e = 1$.*

Proof. The map $\Lambda \rightarrow \text{End}(\Lambda)$, $f \mapsto M_f$, is an injective algebra homomorphism (injective because $M_f(1) = f$). By Theorem 3 it carries p_e to $R_e(-1)$, and $\{p_e\}_{e \geq 1}$ is an algebraically independent generating set of Λ over $\mathbb{Q}(t)$. A polynomial relation among the $R_e(-1)$ would pull back to one among the p_e . For the last clause: p_{de} and $c p_d^e$ are distinct monomials in the free generators whenever $e \geq 2$. $\square$

Remark 6. Corollary 5 is the honest reading of the negative. “ $t = -1$ felt exhausted because $e = 2$ ’s divisor ladder has no interior” was the hope behind Q130. The truth is the opposite: at $t = -1$ the family $\{R_e\}$ is as free as it can possibly be, and the interior of 4 buys nothing, because there is nothing multiplicative to buy.

4 A power-sum expansion of $f_n(t)$, and the coefficient that does the work

Everything in §§5–7 turns on a single coefficient. We isolate it.

Lemma 7. *In $\Lambda[[z]]$,*

$$\sum_{n \geq 1} f_n(t) z^n = \frac{H(z)E(tz) - 1}{1 + t}, \quad H(z) = \sum_{a \geq 0} h_a z^a, \quad E(z) = \sum_{b \geq 0} e_b z^b.$$

Consequently, for $n \geq 1$,

$$f_n(t) = \frac{1}{1+t} \sum_{\nu \vdash n} \frac{1}{z_\nu} \left(\prod_{i=1}^{\ell(\nu)} (1 - (-t)^{\nu_i}) \right) p_\nu. \quad (1)$$

Proof. Pieri gives $h_a e_b = s_{(a,1^b)} + s_{(a+1,1^{b-1})}$ for $a, b \geq 1$, and $h_a = s_{(a)}$. Hence

$$H(z)E(tz) - 1 = \sum_{a \geq 1, b \geq 0} h_a e_b t^b z^{a+b} + \sum_{b \geq 1} e_b t^b z^b.$$

Split the first sum by $b = 0$ and $b \geq 1$ and collect the coefficient of z^n :

$$s_{(n)} + \sum_{b=1}^{n-1} t^b (s_{(n-b,1^b)} + s_{(n-b+1,1^{b-1})}) + t^n e_n = f_n(t) + t(f_n(t) - t^{n-1} s_{(1^n)}) + t^n s_{(1^n)},$$

using $e_n = s_{(1^n)}$; the two $t^n s_{(1^n)}$ terms cancel and the total is $(1+t)f_n(t)$. This is the stated generating function.

For (1), use $H(z) = \exp(\sum_m p_m z^m / m)$ and $E(tz) = \exp(\sum_m (-1)^{m-1} p_m t^m z^m / m)$, so that

$$H(z)E(tz) = \exp\left(\sum_{m \geq 1} \frac{1 - (-t)^m}{m} p_m z^m\right) = \sum_{\nu} \frac{1}{z_\nu} \left(\prod_i (1 - (-t)^{\nu_i}) \right) p_\nu z^{|\nu|},$$

the last step being the standard exponential formula. The subtracted 1 removes only the $n = 0$ term. $\square$

Corollary 8. For $n \geq 1$, the coefficient of p_{1^n} in $f_n(t)$ is $\frac{(1+t)^{n-1}}{n!}$; it vanishes if and only if $t = -1$ and $n \geq 2$. Moreover $f_n(-1) = p_n$.

Proof. Take $\nu = (1^n)$ in (1): $z_\nu = n!$ and each factor is $1 - (-t) = 1 + t$, giving $(1+t)^n / ((1+t)n!)$. For $f_n(-1)$: every factor $1 - (-t)^{\nu_i}$ becomes $1 - 1 = 0$ unless ν_i is odd, and dividing by $1 + t = 0$ requires care, so argue instead from Theorem 3: $f_n(-1) = R_n(-1) \cdot 1 = M_{p_n}(1) = p_n$. $\square$

Remark 9 (the character form). Comparing (1) with $s_\lambda = \sum_{\nu} z_\nu^{-1} \chi_\nu^\lambda p_\nu$ gives the classical hook-character generating function

$$\sum_{k=0}^{n-1} \chi_\nu^{(n-k, 1^k)} t^k = \frac{1}{1+t} \prod_{i=1}^{\ell(\nu)} (1 - (-t)^{\nu_i}), \quad \nu \vdash n,$$

which is how the lemma was checked independently (§9, Test 2). At $t = -1$ it degenerates to χ_ν^λ summed with signs, i.e. to p_n 's hook expansion.

5 The divisor structure survives — as an intertwiner

Theorem 10. For all $d, e \geq 1$,

$$R_{de}(-1) \circ \psi^e = \psi^e \circ R_d(-1)$$

as operators on Λ .

Proof. Let $F \in \Lambda$. By Theorem 3 and the fact that ψ^e is a ring homomorphism with $\psi^e(p_d) = p_{ed}$,

$$\psi^e(R_d(-1)F) = \psi^e(p_d F) = \psi^e(p_d) \psi^e(F) = p_{de} \psi^e(F) = R_{de}(-1)(\psi^e F). \quad \square$$

Corollary 11 (the adjoint, or removal, form). *Let $\varphi_e := (\psi^e)^*$ be the Verschiebung, $\varphi_e(p_{e\mu}) = e^{\ell(\mu)} p_\mu$ and $\varphi_e(p_\nu) = 0$ if some part of ν is not divisible by e . Then*

$$\varphi_e \circ p_{de}^\perp = p_d^\perp \circ \varphi_e.$$

Proof. Adjoin Theorem 10 in the Hall inner product, using $M_f^* = f^\perp$ and $R_e(-1) = M_{p_e}$. The stated formula for φ_e is the adjoint of ψ^e : $\langle \psi^e p_\mu, p_\nu \rangle = z_{e\mu} \delta_{e\mu, \nu}$ and $z_{e\mu}/z_\mu = e^{\ell(\mu)}$. $\square$

So ψ^e conjugates “add an e -ribbon” up to “add a de -ribbon”, and dually φ_e conjugates ribbon removal down. The divisor $d \mid de$ is present — it is just not a power law.

6 Rigidity: the intertwiner detects $t = -1$

The intertwiner of Theorem 10 was proved at $t = -1$. We now show it cannot be deformed — not in t , not in u , not up to a scalar.

Theorem 12. *Let $d \geq 1$, $e \geq 2$, and let c, t, u lie in an extension field of $\mathbb{Q}$ with $c \neq 0$. Suppose*

$$R_{de}(u) \circ \psi^e = c \psi^e \circ R_d(t) \quad (2)$$

as operators on Λ . Then $u = -1$ and $c = 1$; and if $d \geq 2$ then also $t = -1$. Conversely, $u = t = -1$, $c = 1$ satisfies (2).

Proof. The converse is Theorem 10. For the forward direction it suffices to evaluate (2) on the single vector $1 \in \Lambda$, using $\psi^e(1) = 1$:

$$f_{de}(u) = c \psi^e(f_d(t)). \quad (3)$$

Step 1: $u = -1$. The right-hand side of (3) lies in $\text{im } \psi^e = \mathbb{Q}[p_e, p_{2e}, \dots]$, whose degree- de component is spanned by $\{p_{e\nu} : \nu \vdash d\}$. Since $e \geq 2$, the partition (1^{de}) is not of the form $e\nu$, so the coefficient of $p_{1^{de}}$ on the right is 0. By Corollary 8 that coefficient on the left is $(1+u)^{de-1}/(de)!$. As $de \geq 2$, this forces $u = -1$.

Step 2: $t = -1$ and $c = 1$. With $u = -1$, Corollary 8 gives $f_{de}(-1) = p_{de} = p_{e \cdot (d)}$. Expand the right-hand side by (1):

$$c \psi^e(f_d(t)) = \frac{c}{1+t} \sum_{\nu \vdash d} \frac{1}{z_\nu} \left(\prod_i (1 - (-t)^{\nu_i}) \right) p_{e\nu}.$$

The partitions $\{e\nu : \nu \vdash d\}$ are pairwise distinct, so $\{p_{e\nu}\}_{\nu \vdash d}$ is linearly independent, and (3) may be read coefficientwise.

Suppose $d \geq 2$. Then $(1^d) \neq (d)$, and the coefficient of $p_{e \cdot (1^d)}$ must vanish; by Corollary 8 it equals $c(1+t)^{d-1}/d!$. Since $c \neq 0$ this forces $t = -1$. Then $f_d(-1) = p_d$, so $\psi^e(f_d(t)) = p_{de}$ and matching the $p_{e \cdot (d)}$ coefficients gives $c = 1$.

If $d = 1$ then $f_1(t) = s_{(1)} = p_1$ carries no t , and $\psi^e(p_1) = p_e$; matching (3), which now reads $p_e = c p_e$, gives $c = 1$. $\square$

Remark 13 (what is new here). Q75 already characterised $t = -1$ as the unique point where $R_e(t)$ is a multiplication operator. Theorem 12 is a different characterisation, and the obstruction is a different one: it is the p_{1^n} -coefficient $(1+t)^{n-1}/n!$, i.e. the pairing of $f_n(t)$ against the *regular representation*. The two are compatible — both single out $t = -1$ — but neither is formally a restatement of the other, since Theorem 12 allows an *independent* reparametrisation u of the target and a scalar c , and still leaves no freedom. In particular the natural guesses “ $u = t^e$ ” or “ $c = t^{\text{something}}$ ” are all excluded.

Remark 14 (degenerate cases, stated). The hypothesis $e \geq 2$ is necessary: for $e = 1$, $\psi^1 = \text{id}$ and (2) reads $R_d(u) = c R_d(t)$, satisfied by $c = 1$, $u = t$ arbitrary. So the rigidity is genuinely a statement about *proper* divisors, and it does not hold vacuously.

7 The sharp diagnosis: ψ^e divides ribbon size

Theorem 10 says ψ^e carries the d -ribbon operator to the de -ribbon operator. Read in the other direction: *along ψ^e , ribbon size is divided by e* . This makes the failure of Q130 visible in one line.

Proposition 15. $R_2(-1)^2 \circ \psi^2 = \psi^2 \circ R_1(-1)^2$. Consequently, if $R_4(-1) = c R_2(-1)^2$ for a scalar c , then $p_2 = c p_1^2$.

Proof. The first identity is Theorem 10 with $(d, e) = (1, 2)$, applied twice: $R_2(-1)\psi^2 = \psi^2 R_1(-1)$, hence $R_2(-1)^2 \psi^2 = R_2(-1)\psi^2 R_1(-1) = \psi^2 R_1(-1)^2$.

Now assume $R_4(-1) = c R_2(-1)^2$. Compose on the right with ψ^2 and use Theorem 10 with $(d, e) = (2, 2)$ on the left:

$$\psi^2 \circ R_2(-1) = R_4(-1) \circ \psi^2 = c R_2(-1)^2 \circ \psi^2 = c \psi^2 \circ R_1(-1)^2.$$

ψ^2 is injective, so $R_2(-1) = c R_1(-1)^2$; evaluating at 1 gives $p_2 = c p_1^2$. □

This is the whole error, exhibited. Upstairs, R_4 is *one* ribbon of size 4; downstairs it is *one* domino. But $R_2(-1)^2$ is *two* dominoes upstairs, and downstairs it is *two boxes*. The ladder $Q'_{(r^l)} = (Q'_{(r^d)})^{l/d}$ is a statement about how many times a fixed vector is multiplied by itself; the transcription reused that exponent as the number of times an *operator* is composed, and those are different indices. A divisor relation between ribbon sizes is a relation between *one* ribbon and *one* ribbon — a conjugation — never between one ribbon and many.

Remark 16. Proposition 15 also explains why the failure looked so violent in the data: the ratio of matrix entries of $R_4(-1)$ to those of $R_2(-1)^2$ takes all three values $-1, 0, +1$ in every degree checked (§9, Test 0). A 0 appears exactly where p_2^2 has support that p_4 does not, e.g. at $s_{(2,2)}$ — the shape R_4 cannot reach because it is not a border strip. “Two dominoes” can build a 2×2 square; “one 4-ribbon” cannot. That 2×2 square is the geometric form of the obstruction.

8 By-product: the cocycle exponent at even e

The brief asked, for a different question (Q121), that the competing exponents t^{e-1} (mine, proved) and t^2 (an agent’s derivation attributed to Graf, 2511.01114) be separated at $e = 4$, since they agree identically at $e = 3$. The work above hands this over for free.

Proposition 17. *Specialising the proved cocycle $\omega R_e(t)\omega = t^{e-1}R_e(1/t)$ at $t = -1$ and using Theorem 3 gives*

$$\omega M_{p_e} \omega = (-1)^{e-1} M_{p_e},$$

i.e. the classical $\omega(p_e) = (-1)^{e-1}p_e$. In particular the exponent $e - 1$ is confirmed by a fact fitted to nothing, and $t^{e-1}|_{t=-1} = (-1)^{e-1}$ differs from $t^2|_{t=-1} = +1$ for every even e , in particular at $e = 4$.

Proof. Set $t = -1$ in the cocycle: $\omega R_e(-1)\omega = (-1)^{e-1}R_e(-1)$. By Theorem 3 both sides are multiplication operators; evaluate at 1, using $\omega(1) = 1$, to get $\omega(p_e) = (-1)^{e-1}p_e$, which is classical. The numerical claim is immediate. $\square$

Remark 18 (the caveat, and it is load-bearing). Proposition 17 refutes t^2 as a cocycle for ω acting on my $R_e(t)$. It does not refute anything Graf wrote. The brief itself flagged the risk: if Graf's α_z, β_z carry no ribbon-size index, then an e -independent exponent t^2 is e -independent *by construction* and comparing it to t^{e-1} is a category error, not a contest. I had no network access this session and did not read 2511.01114; the antecedent is unchecked. The honest statement is therefore conditional, and the e -independence of t^2 is itself weak evidence for the category-error reading. This is recorded as a gap in §10 and *not* as a refutation of a peer's result.

9 Computational verification

All code is at `probes/2026-09-09-c2-Q130/` (`ribbon.py`, `symfunc.py`, `q130_test.py`, `verify.py`, `verify2.py`). Ribbon operators are built *from the border-strip definition* (connected, no 2×2 square), and the Schur $\leftrightarrow$ power-sum transition is built from an independent Murnaghan–Nakayama character recursion, so Test 1 is a genuine check of Theorem 3 and not a restatement of it.

Test 0 $R_4(-1)$ against $R_2(-1)^2$, as matrices from degree n to $n + 4$ for $n = 0, \dots, 4$ (up to 42 nonzero cells). The set of entrywise ratios is $\{-1, 0, +1\}$ in every degree, so no scalar c exists. **Refuted.**

Test 1 $R_e(-1)s_\mu = p_e \cdot s_\mu$ for $e = 1, \dots, 5$ and all μ with $|\mu| + e \leq 9$. **Pass.**

Test 2 The hook-character identity $\sum_k \chi_\nu^{(n-k, 1^k)} t^k = \prod_i (1 - (-t)^{\nu_i}) / (1 + t)$ for all ν with $|\nu| \leq 8$, symbolically in t . **Pass.** (Independent check of Lemma 7.)

Test 3 The two rigidity coefficients of Theorem 12, computed symbolically for $(d, e) \in \{(2, 2), (2, 3), (3, 2), (1, 4), (2, 4)\}$ $[p_{1de}]f_{de}(u) = (1 + u)^{de-1}/(de)!$ and $[p_{ed}]\psi^e f_d(t) = (1 + t)^{d-1}/d!$, both nonzero away from -1 , and the second identically 1 when $d = 1$. **Pass.**

Test 4 The intertwiner $R_{de}(-1)\psi^e = \psi^e R_d(-1)$ on 44 instances (d, e, μ) with (d, e) ranging over nine pairs including $(4, 2)$ and $(2, 4)$. **Pass.**

Test 5 The cocycle $\omega R_e(t)\omega = t^{e-1}R_e(1/t)$ symbolically in t , $e = 1, \dots, 5$, $|\mu| + e \leq 9$. **Pass**, re-confirming the Q104 result on a fresh implementation.

Test 6 The descent $R_2(-1)^2\psi^2 = \psi^2 R_1(-1)^2$ of Proposition 15. **Pass.**

Test 7 The adjoint form $\varphi_e p_{de}^\perp = p_d^\perp \varphi_e$ of Corollary 11, 402 instances. **Pass.**

Test 8 The p -expansion (1) of $f_n(t)$ for $n \leq 8$, symbolically. **Pass.**

Remark 19 (what varies, stated before the fact). Per my standing rule that a control must move a prediction: Test 0’s discriminating variable is the pair $(e, \text{composite structure})$, and it does move — the two operators differ in *every* degree tested and the difference is not a scalar. Test 4’s discriminating variable is (d, e) with de fixed: the pairs $(4, 2)$ and $(2, 4)$ send the *same* $R_8(-1)$ through *different* Adams operations, and both pass, which is what an intertwining (as opposed to a coincidence at one factorisation) predicts. Test 5 varies e across the parity that separates t^{e-1} from t^2 .

10 Gaps, omissions, and stale premises

1. **The rival exponent is unchecked at source.** Proposition 17 is conditional; see Remark 18. Reading 2511.01114 to determine whether α_z, β_z carry a ribbon-size index is the outstanding action. Until then t^2 is *not* refuted — only shown incompatible with a cocycle for ω on R_e .
2. **The $t = 2$ control was not run**, deliberately. The brief made it mandatory; the evening browse log, written afterwards, retired it (“its premise is refuted; running it would produce a green check on a structure that does not exist”) and reinstated it only behind a re-derivation of $1 - Q = (1 - \beta)(1 - q^2)$ that I have not performed. Running a control whose premise is known-refuted would have produced exactly the kind of vacuous green check I have been burned by before. It remains owed, gated on that re-derivation.
3. **Q121 was not attempted.** The gate in the brief directed me to the later log, which ranked Q130 first. Zabrocki’s flip $\hat{V}$ is untouched. Note also that the brief asserts `math/0008188` sits at `agent-summary`; it is at `deep-read` in `sources.json` as of this session. The brief was written before that update.
4. **The source ladder is not verified by me.** Descouens–Morita `math/0611522` Prop. 6.1 is recorded at `deep-read` but the read was an agent’s. Nothing here depends on it; §7 refutes the operator transcription only. Whether the source statement itself is correctly recorded remains open.
5. **Not proved: a t -deformed intertwiner for some other map.** Theorem 12 shows no deformation exists *with ψ^e fixed*. Whether some t -dependent family Ψ_t^e with $\Psi_{-1}^e = \psi^e$ intertwines $R_{de}(t)$ with $R_d(t)$ is open, and is the natural next question — the e -quotient map of LLT theory is the obvious candidate.

11 Summary

Q130 is decided in the negative, and decided structurally: at $t = -1$ the ribbon operators are multiplication by power sums, hence algebraically independent, hence admit no multiplicative ladder of any shape. What the divisor $d \mid de$ does give is an intertwining by the Adams operation ψ^e , and that intertwining is rigid — it detects $t = -1$ exactly, with no freedom in the parameter, in a second parameter, or in a scalar. The transcription failed because ψ^e divides *ribbon size*, not *ribbon count*, and a power law confuses the two.